

TalentForge POC — 2-Person, 4-Week Execution Plan

Team: Karthikeyan (Senior Lead / Frontend Architect + Tool Layer) · Rohan (Fullstack / Backend & Infra) **Start:** Monday, July 13, 2026 · **Launch:** Friday, August 14, 2026 **Goal:** A customer-ready CSE challenge platform — students solve problems in a sandboxed editor, get auto-graded in <5s, earn verified badges, and employers discover them.

1. Scope Decisions (locked for this timeline)

Area	In Scope (Aug 14)	Deferred (Post-launch)
Domain	CSE only	ECE / SPICE simulation
Languages	Python 3.11, Node 18 (JS), Java 17	C++, Go, Rust
Problems	5-8 curated problems incl. flagship "Build a Load Balancer"	Full 5-tier catalog
Auth	Email/password + Google OAuth	GitHub, Microsoft OAuth
Grading	Correctness (60%) + complexity heuristic (30%) + style/lint (10%)	ML-based grading, MOSS plagiarism
Psychometric	15-min short-form assessment (Big Five lite + logic)	Adaptive 30-min engine, XGBoost matching
Badges	DB-verified badge + PDF/PNG certificate + public verify URL	—
Blockchain	ERC-721 on Polygon Amoy testnet (stretch, Week 4)	Mainnet, IPFS metadata, Torus wallet

Area	In Scope (Aug 14)	Deferred (Post-launch)
Expert review	Simple reviewer role + approve/reject + comment	Full expert marketplace, payments, SLA engine
Employer	Discovery list, filters, candidate profile, shortlist	Messaging, pipeline analytics, matching engine
Infra	Single VPS or EC2 t3.medium + Docker Compose, Postgres, Redis, S3	Kubernetes autoscaling, Datadog

Repo structure: npm workspaces monorepo — `/backend` (Node.js + Express + TypeScript + Prisma), `/frontend` (React + Vite + Tailwind + Monaco), `/worker` (grading workers), `/contracts` (Hardhat, stretch).

2. Ownership Split

Karthikeyan — Tool & Experience Layer

- Monaco editor workspace, run/submit flow, live results (socket.io client)
- Student dashboard, problem board, badge/certificate UI
- Employer discovery portal + reviewer UI
- LLM-powered feedback formatter (turn raw grader output into actionable coaching text — leverages LangChain/Ollama experience)
- Landing page, onboarding, responsive polish, Lighthouse/accessibility

Rohan — Platform & Execution Layer

- Express API, Prisma schema, Postgres, Redis, BullMQ
- Auth (JWT + Google OAuth), S3 uploads

- Docker sandbox (network-disabled, 30s timeout, 256MB RAM, read-only FS, seccomp)
- Grading engine (correctness runner, complexity estimator, ESLint/pylint/checkstyle)
- CI/CD, deployment, monitoring (Sentry), backups

Shared contract: OpenAPI spec + WebSocket event schema frozen by end of Week 1, Day 3. Frontend develops against a mock server until real endpoints land.

3. Week-by-Week Plan

WEEK 1 (Jul 13-19) — Foundation & Contracts

Sprint goal: Repos live, DB migrated, auth working, editor rendering, API contract frozen.

Day	Karthikeyan	Rohan
Mon-Tue	Scaffold <code>/frontend</code> : Vite + React 18 + TS + Tailwind + React Router + TanStack Query. Design tokens, layout shell, auth pages UI.	Scaffold monorepo + <code>/backend</code> : Express + TS + Prisma. Docker Compose (Postgres 16, Redis 7). Prisma schema: <code>User</code> , <code>Problem</code> , <code>Submission</code> , <code>Badge</code> , <code>PsychProfile</code> , <code>Review</code> . Run first migration.
Wed	Joint (2h) : Freeze OpenAPI spec (~25 endpoints) + socket.io events <code>(grading:queued/running/complete/failed)</code> . K sets up MSW mock server from spec.	Auth: <code>/auth/register</code> , <code>/auth/login</code> , <code>/auth/me</code> , refresh tokens (Redis), bcrypt, Google OAuth (passport).
Thu	Monaco Editor integration: language selector (py/js/java), themes, syntax highlight, boilerplate templates per problem.	S3 (or MinIO locally) upload service, presigned URLs. <code>Problem</code> CRUD + seed script with 3 problems incl. test cases (hidden).
Fri	Problem board page + problem detail page (statement, constraints, examples) wired to mock API. Auth flow wired end-to-end against real backend.	BullMQ setup: <code>grading</code> queue, worker skeleton in <code>/worker</code> , job lifecycle events → socket.io emit. GitHub Actions CI (lint + test + build). Deploy staging (single EC2/VPS, Docker Compose, Caddy/Nginx + SSL).

Exit criteria: Register → login → browse problems → open editor works on staging. CI green. Contract frozen.

WEEK 2 (Jul 20-26) — Sandbox Execution & Grading (Critical Path)

Sprint goal: Code runs safely in Docker, gets graded, score returned live to the browser.

Day	Karthikeyan	Rohan
Mon	Run/Submit UX: submit button → optimistic "queued" state → socket.io live status → results panel skeleton.	Sandbox images: <code>python:3.11-slim</code> , <code>node:18-alpine</code> , <code>eclipse-temurin:17</code> runner images with harness scripts. Enforce: <code>--network=none</code> , <code>--memory=256m</code> , <code>--read-only</code> , tmpfs <code>/tmp</code> , 30s kill timer, non-root user, seccomp profile.
Tue	Results dashboard: score ring (0–100), per-test-case pass/fail table (expected vs actual), execution time & memory per case.	Correctness runner: execute against hidden test cases, capture stdout/stderr/exit code, diff outputs, compute correctness %.
Wed	Feedback panel: render grader feedback; build LLM feedback formatter (Claude API or local model via backend proxy) that converts raw results into 3 coaching bullets. Feature-flag it.	Complexity heuristic (input-size scaling runs: n , $2n$, $4n \rightarrow$ fit curve) + style scoring: pylint / ESLint / checkstyle in-container, map to 0–100. Composite score: 60/30/10. Persist to <code>Submission</code> .
Thu	Submission history page (attempts, scores, diffs of score over time). Resubmit flow (cooldown configurable, default 24h for POC).	Static pre-check: block dangerous patterns (<code>subprocess</code> , <code>os.system</code> , <code>child_process</code> , <code>Runtime.exec</code>), file/net syscalls) before execution. Graceful failure paths: timeout, OOM, compile error → clear user-facing messages, queue never blocks.
Fri	Joint: End-to-end test — 10 sample solutions (correct, wrong, slow, malicious, infinite loop) across all 3 languages. Fix everything found.	Load test: 20 concurrent submissions, grading $p95 < 5s$. Worker concurrency + container pooling tuning. Sentry wired (backend + worker).

Exit criteria: Any of the 3 languages: submit → graded → live score in browser in <5s.
Malicious code safely rejected. Zero queue deadlocks.

WEEK 3 (Jul 27-Aug 2) — Badges, Psychometric, Reviewer & Employer Portal

Sprint goal: Full student loop closes (badge earned); employer can discover candidates.

Day	Karthikeyan	Rohan
Mon	Badge award UI: celebration modal on score ≥ 75 , badge gallery on profile, shareable public verify page (<code>/verify/:badgeId</code>) with score, problem, date, expert status.	Badge service: auto-award on score ≥ 75 , unique verify ID, signed verification endpoint. Certificate generation (PDF via Puppeteer or pdf-lib) with QR to verify URL.
Tue	Psychometric assessment UI: 15-min timed short-form (Big Five lite ~30 items + 10 logic questions), progress bar, results radar chart.	Psychometric API: question bank seed, scoring engine (trait indices per PRD weights: Logical 25 / Detail 20 / Learning 20 / Persistence 20 / Communication 15), persist <code>PsychProfile</code> .
Wed	Reviewer UI: submission queue, code viewer with syntax highlighting, 1-5 star rating, comment box, approve/reject for badge.	Reviewer role + RBAC middleware. Review endpoints, assignment logic (oldest-first queue). Review outcome updates badge status (AI-provisional $\rightarrow$ expert-verified).
Thu	Employer portal: register/login as employer, candidate list with filters (badge earned, score $> X$, language, college), candidate detail (scores, badges, code samples, psych radar), shortlist button.	Employer endpoints: candidate search/filter (indexed queries), shortlist model, code-sample read access controls (students opt-in flag). Rate limiting (express-rate-limit + Redis).

Day	Karthikeyan	Rohan
Fri	LinkedIn share (badge OG image + share URL), landing page v1 with real product screenshots, onboarding tour.	Seed 5 more problems (total 8) incl. flagship " Build a Load Balancer " with 10 hidden test cases + rubric. DB backups (nightly pg_dump → S3). Staging → production environment split.

Exit criteria: Student loop: register → psychometric → solve → score ≥ 75 → badge → verify URL → visible to employer → shortlisted. All on staging.

WEEK 4 (Aug 3-9) — Hardening, Beta & Blockchain Stretch

Sprint goal: 15-25 beta users on the platform; production-grade stability.

Day	Karthikeyan	Rohan
Mon	Mobile responsiveness pass (editor read-only on mobile is acceptable; dashboards fully responsive). Empty states, error states, loading skeletons everywhere.	Security hardening: helmet, CORS lockdown, input validation (zod) on all endpoints, SQLi/XSS sweep, JWT expiry/rotation audit, secrets to env vault.
Tue	Beta launch (invite 15-25 students from your network/college contacts). Monitor Discord/WhatsApp feedback channel. Hotfix UI issues same-day.	Monitor Sentry + logs live. Hotfix backend issues. Query optimization + indexes for candidate search and submission history.
Wed	Beta feedback triage with Rohan (joint, 1h). Fix top 5 UX issues. Lighthouse pass: target >90 performance, page load <1.5s.	Load test 100 concurrent users (k6 or artillery). Verify autos-restart policies, disk alerts, Redis persistence. Fix top 5 backend issues.
Thu	Stretch: Torus/web3 wallet-less claim UI — "Mint badge on-chain" button, tx status, PolygonScan link. Falls back gracefully if contracts not ready.	Stretch: Hardhat project in <code>/contracts</code> : minimal ERC-721 (OpenZeppelin), deploy to Polygon Amoy, backend mint endpoint (server-side signer wallet, gas-sponsored). If behind schedule: skip — DB badges ship.
Fri	Final copy editing, FAQ page, 2-min demo video (screen recording). Employer one-pager PDF.	Production runbook (what to do when queue stalls / DB down / sandbox breach suspected). Final backup + restore drill. Go/no-go checklist review (joint).

Exit criteria: Beta NPS informally positive, zero P1 bugs, production environment stable under 100 concurrent users.

LAUNCH WEEK (Aug 10-14) — Go Live

Day	Both
Mon Aug 10	Code freeze (bugfixes only). Full regression of all critical flows on prod. DNS + SSL verified on final domain.
Tue Aug 11	Soft launch: open signups to ~50 invited students. Verify signup→grade→badge on production with real users.
Wed Aug 12	Fix anything soft launch surfaced. Warm outreach to 5-10 employer/HR contacts with the one-pager; get 3+ employer accounts created.
Thu Aug 13	Final DB backup. Monitoring alerts confirmed (Sentry + uptime ping). Prepare launch posts (LinkedIn founder post, college groups, Discord).
Fri Aug 14	 Public launch. Publish posts 10 AM IST. Both on-call all day. Track: signups, submissions, grading success rate, badge awards, employer views. Evening: metrics snapshot for investor/customer conversations.

Launch targets (realistic for 2-person POC): 100+ signups, 30+ submissions, 15+ badges, 3-5 employer accounts, 1+ shortlist, 99%+ uptime, grading p95 <5s.

4. Tech Stack Summary (per component)

Component	Stack	Owner
Frontend	React 18, Vite, TypeScript, Tailwind CSS, Monaco Editor, TanStack Query, socket.io-client, Recharts (psych radar/score charts)	Karthikeyan

Component	Stack	Owner
Backend API	Node.js 20, Express, TypeScript, Prisma, zod, passport (Google OAuth), jsonwebtoken, bcrypt	Rohan
Data	PostgreSQL 16 (primary), Redis 7 (sessions, BullMQ), AWS S3 (submissions, certificates)	Rohan
Grading workers	BullMQ, dockerode, per-language Docker runner images (python:3.11-slim, node:18-alpine, temurin:17), pylint/ESLint/checkstyle	Rohan
Real-time	socket.io (server: Rohan, client: Karthikeyan)	Shared
LLM feedback	Anthropic API (or Ollama local) via backend proxy, feature-flagged	Karthikeyan
Blockchain (stretch)	Solidity, OpenZeppelin ERC-721, Hardhat, Polygon Amoy, ethers.js server-side signer	Rohan (contract) / Karthikeyan (UI)
DevOps	GitHub Actions CI, Docker Compose on EC2 t3.medium (or Hetzner/DO), Caddy (auto-SSL), Sentry, UptimeRobot, nightly pg_dump→S3	Rohan

Estimated infra cost: ₹8,000–15,000/month (one VM + RDS-lite/self-hosted PG + S3 + Sentry free tier) — a fraction of the original ₹75K/month plan, appropriate for POC scale.

5. Key Risks (2-person edition)

1. **Sandbox is the critical path.** If Week 2 slips, cut Java (ship Python + JS only). Never cut the security constraints.

2. **Bus factor = 1 per layer.** Mitigate: daily 15-min sync, shared PRs (each reviews the other's code), README-driven setup so either can run the whole stack.
 3. **Blockchain eats time.** It's explicitly a stretch. DB-verified badges + public verify URL deliver the customer value; on-chain is a demo bonus.
 4. **Scope creep from PRD.** ECE, tiers, matching engine, MOSS, payments — all post-launch. Say no until Aug 15.
 5. **Beta recruiting.** Start collecting 25 committed beta students in Week 1 (not Week 4) via college/HOD contacts.
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6. Working Agreements

- Daily standup 9:30 AM IST (15 min): yesterday / today / blockers.
- All work via PRs; cross-review within 12 hours.
- Friday 4 PM demo to each other + any stakeholder; retro 30 min.
- Trunk-based with short-lived branches; staging auto-deploys on merge to `main`; prod deploys are manual and tagged.